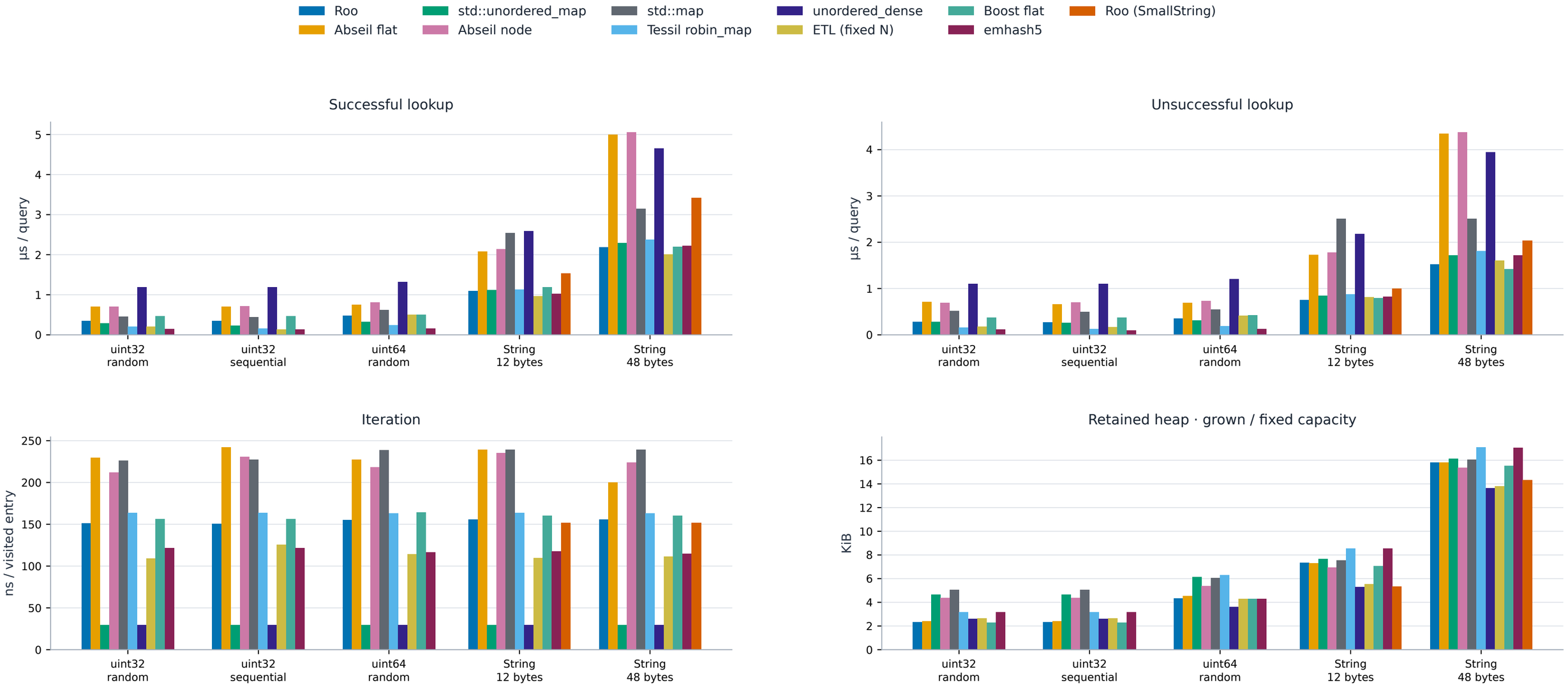
Current map comparison · 128 entries

Default hashers; std::map uses ordering. Heap includes the map object and allocator overhead; it is not peak memory.



Roo fill limit 75% · ESP32 · 240 MHz · -std=gnu++17 -O2 -DNDEBUG · warm access · heap checks: light · 1 suite/case, 7 samples/row

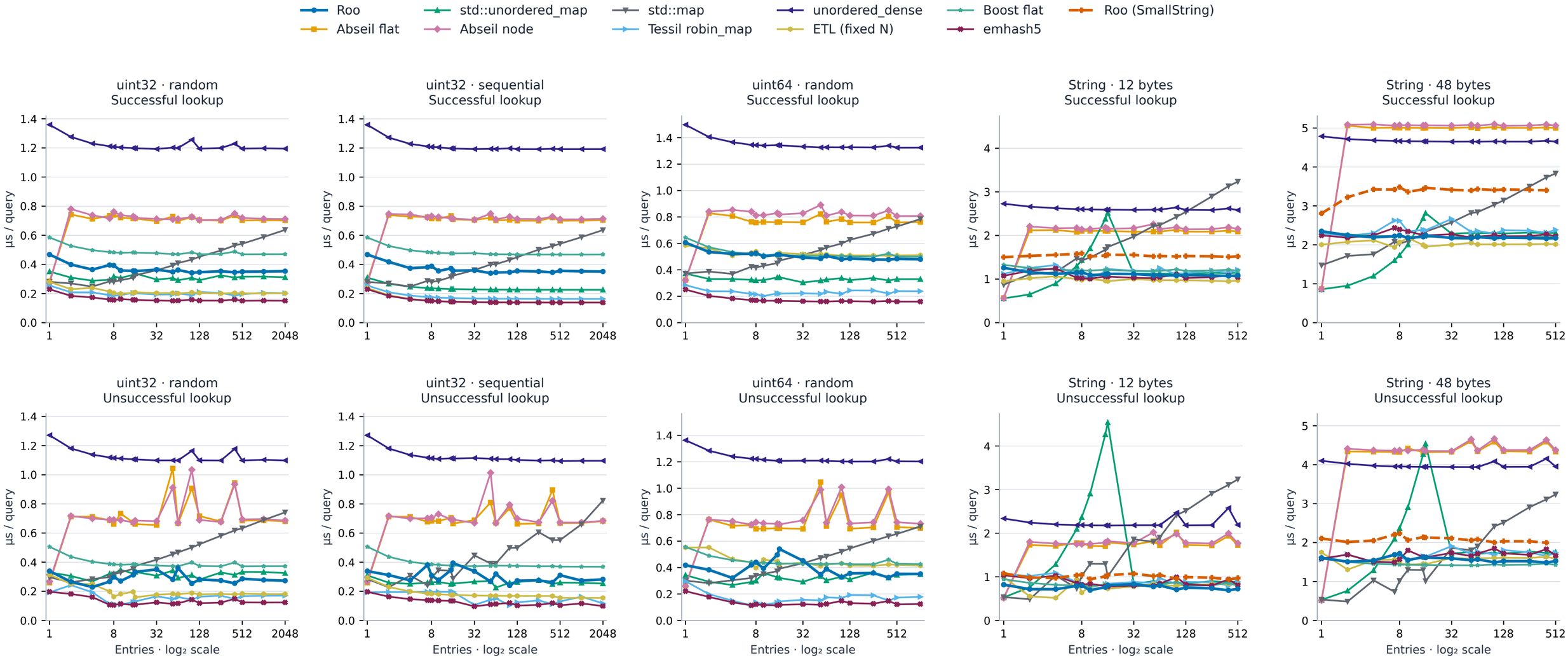
Medians shown; no confidence intervals. Default capacity policies; fill ratios are not normalized. Lower is better.

ETL: fixed capacity N, N buckets; embedded storage + adapter included; no dynamic-growth series.

Roo (SmallString): separate capture/board; other string series use std::string. 48-character SmallString ends at 400 entries.

Lookup performance across map sizes

Default hashers; std::map uses ordering. Y axes are linear and start at zero.



Roo fill limit 75% · ESP32 · 240 MHz · -std=gnu++17 -O2 -DNDEBUG · warm access · heap checks: light · 1 suite/case, 7 samples/row

Medians shown; no confidence intervals. Default capacity policies; fill ratios are not normalized. Lower is better.

ETL: fixed capacity N, N buckets; embedded storage + adapter included; no dynamic-growth series.

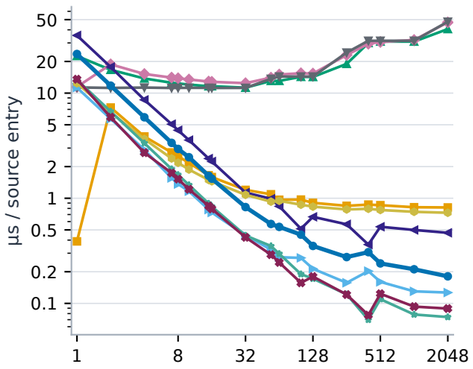
Roo (SmallString): separate capture/board; other string series use std::string. 48-character SmallString ends at 400 entries.

Copy and construction costs

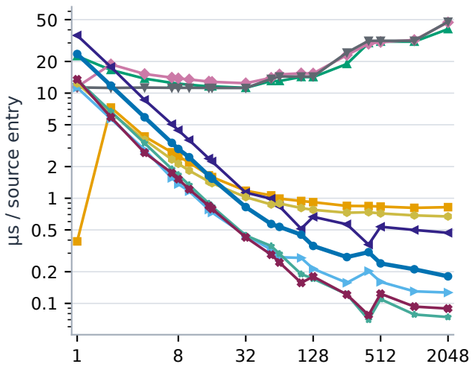
Default hashers; std::map uses ordering. Y axes are logarithmic.

Roo · Abseil flat · std::unordered_map · Abseil node · std::map · Tessil robin_map · unordered_dense · ETL (fixed N) · Boost flat · emhash5 · Roo (SmallString)

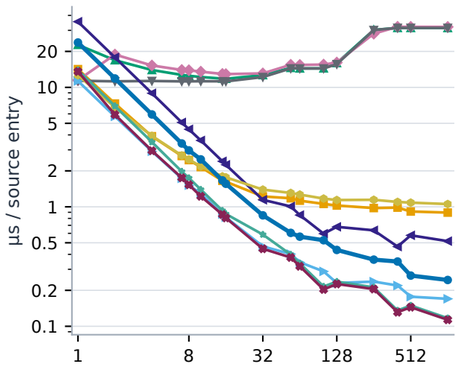
uint32 · random
Copy + destruction



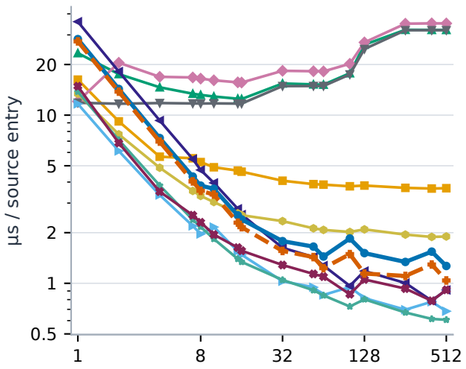
uint32 · sequential
Copy + destruction



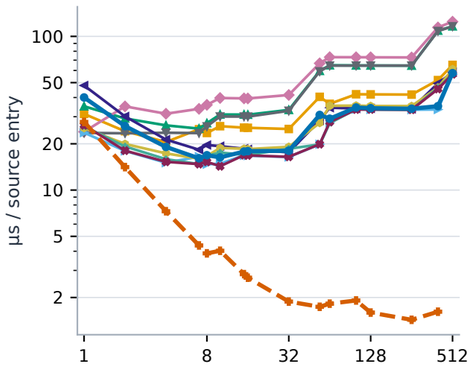
uint64 · random
Copy + destruction



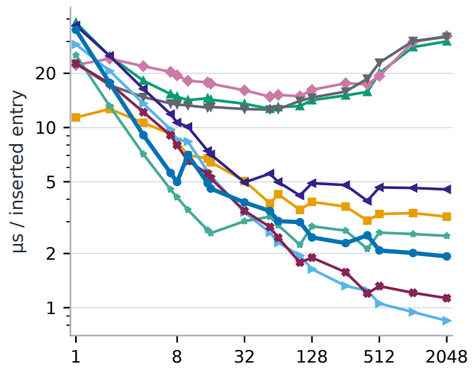
String · 12 bytes
Copy + destruction



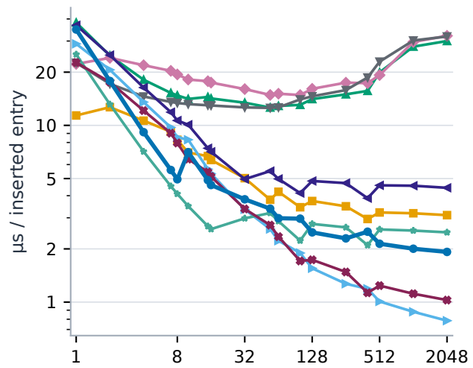
String · 48 bytes
Copy + destruction



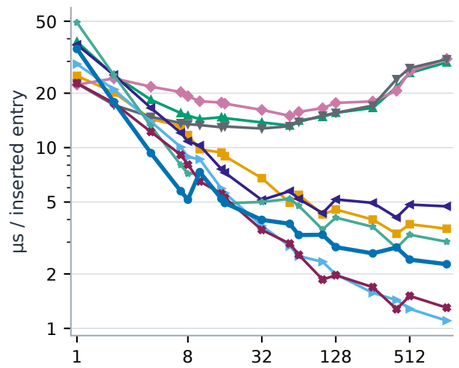
uint32 · random
Build with growth + destruction



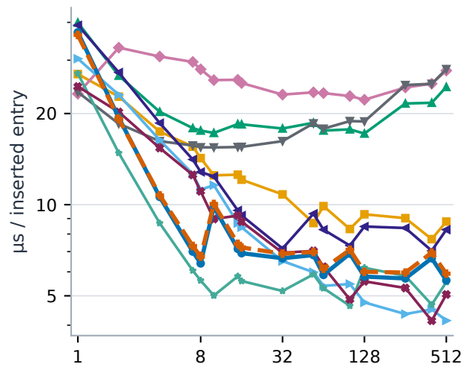
uint32 · sequential
Build with growth + destruction



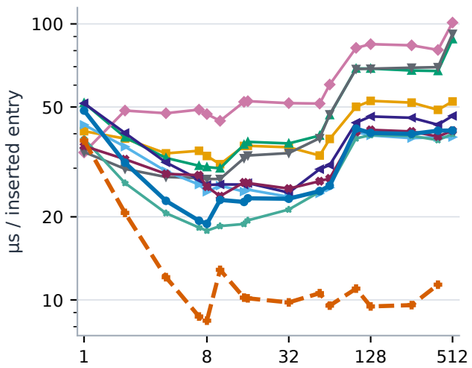
uint64 · random
Build with growth + destruction



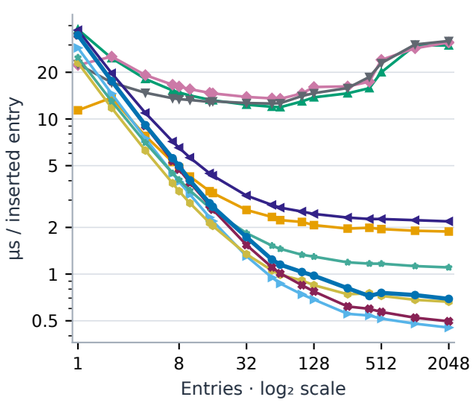
String · 12 bytes
Build with growth + destruction



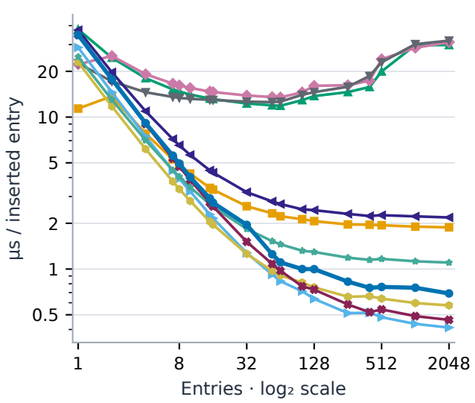
String · 48 bytes
Build with growth + destruction



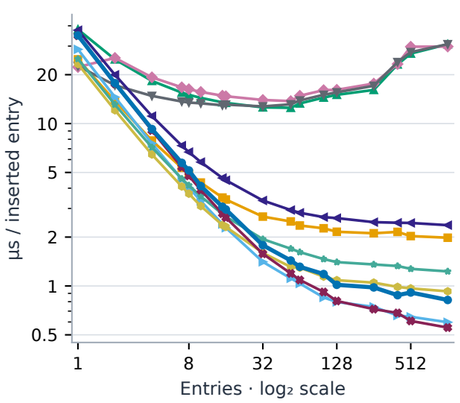
uint32 · random
Presized build + destruction



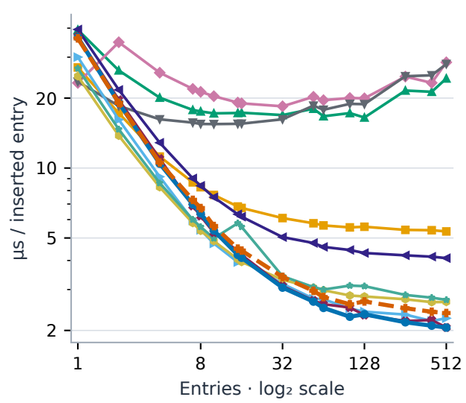
uint32 · sequential
Presized build + destruction



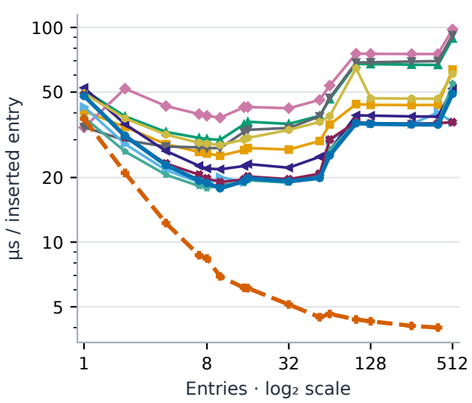
uint64 · random
Presized build + destruction



String · 12 bytes
Presized build + destruction



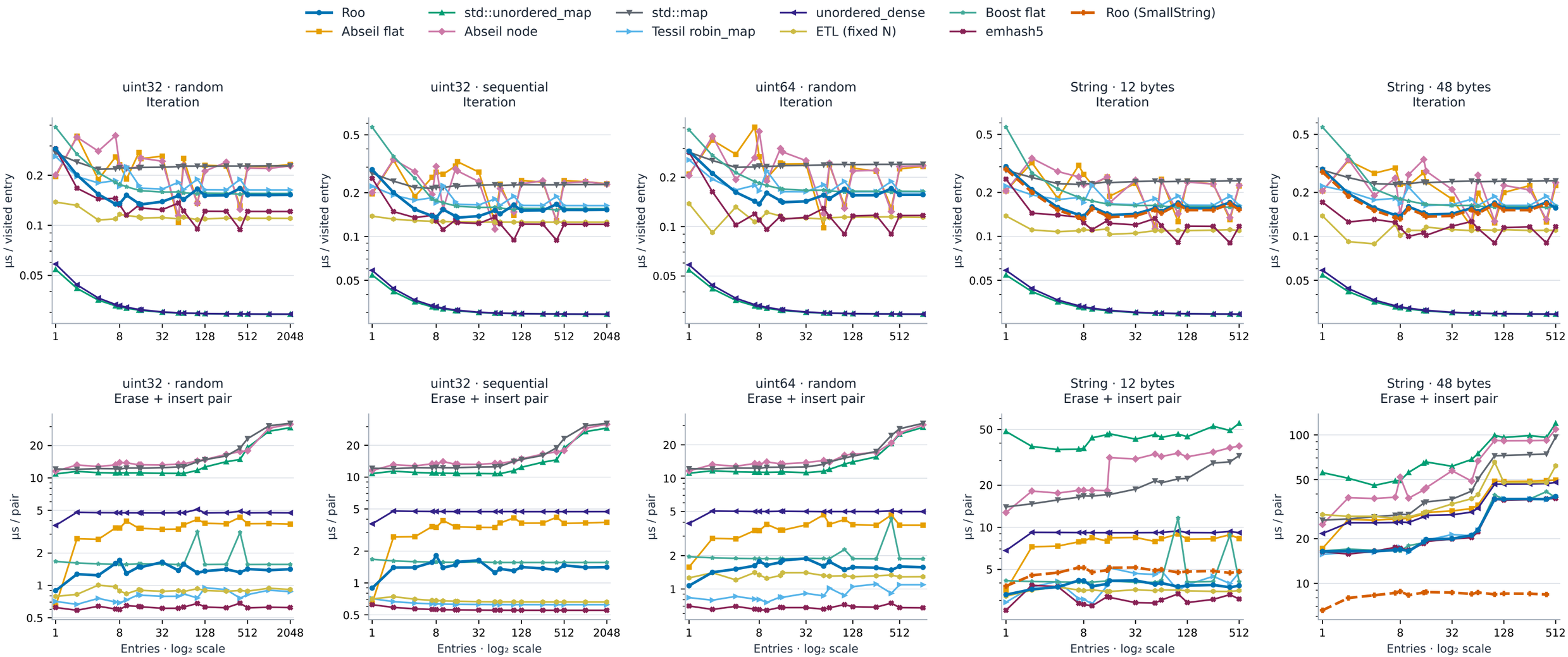
String · 48 bytes
Presized build + destruction



Roo fill limit 75% · ESP32 · 240 MHz · -std=gnu++17 -O2 -DNDEBUG · warm access · heap checks: light · 1 suite/case, 7 samples/row
Medians shown; no confidence intervals. Default capacity policies; fill ratios are not normalized. Lower is better.
ETL: fixed capacity N, N buckets; embedded storage + adapter included; no dynamic-growth series.
Roo (SmallString): separate capture/board; other string series use std::string. 48-character SmallString ends at 400 entries.

Iteration and erase/insert costs

Default hashers; std::map uses ordering. Y axes are logarithmic.



Roo fill limit 75% · ESP32 · 240 MHz · -std=gnu++17 -O2 -DNDEBUG · warm access · heap checks: light · 1 suite/case, 7 samples/row

Medians shown; no confidence intervals. Default capacity policies; fill ratios are not normalized. Lower is better.

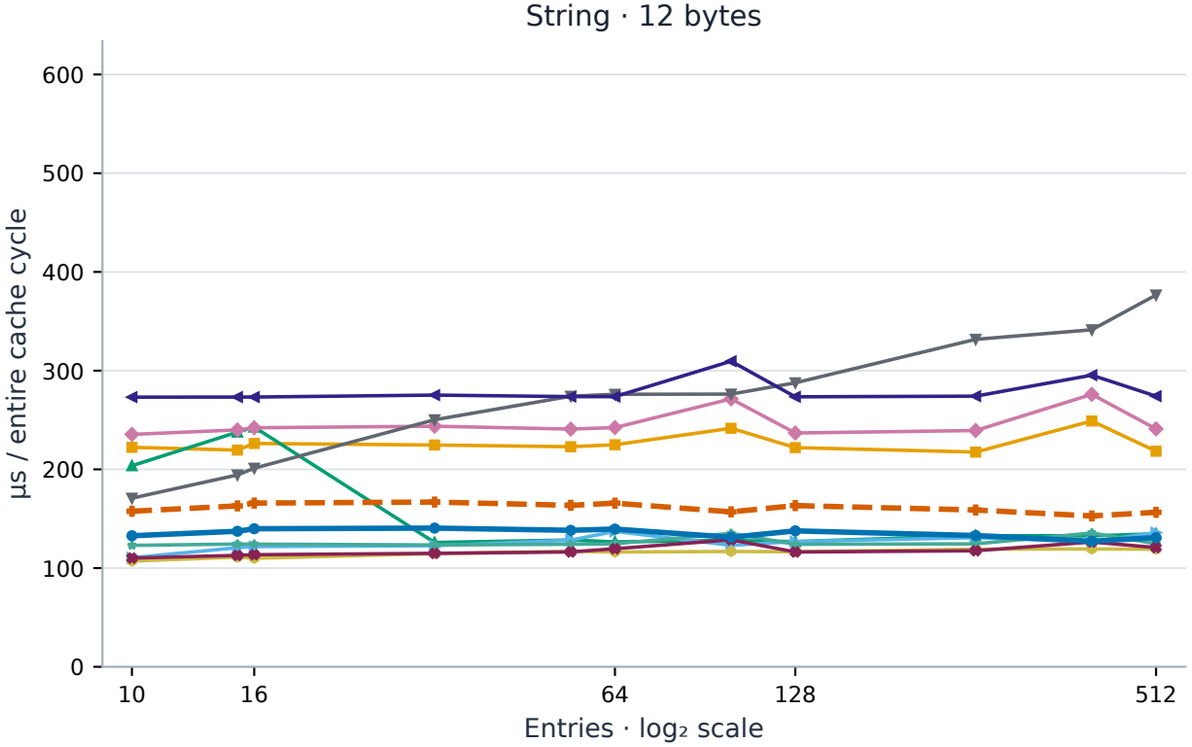
ETL: fixed capacity N, N buckets; embedded storage + adapter included; no dynamic-growth series.

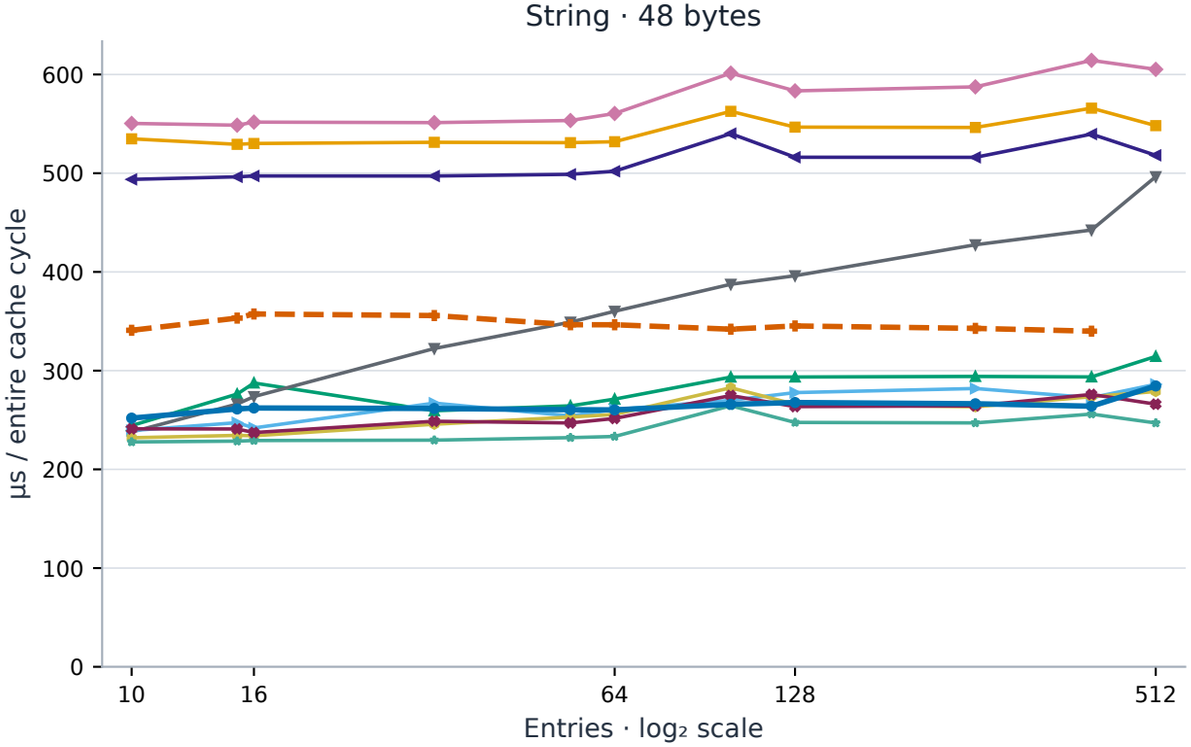
Roo (SmallString): separate capture/board; other string series use std::string. 48-character SmallString ends at 400 entries.

Cache churn · string keys

Default hashers; std::map uses ordering. One cycle = 90 hits + 10 misses + one eviction + one insertion.

Roo · Abseil flat · std::unordered_map · Abseil node · std::map · Tessil robin_map · unordered_dense · ETL (fixed N) · Boost flat · emhash5 · Roo (SmallString)





Roo fill limit 75% · ESP32 · 240 MHz · -std=gnu++17 -O2 -DNDEBUG · warm access · heap checks: light · 1 suite/case, 7 samples/row

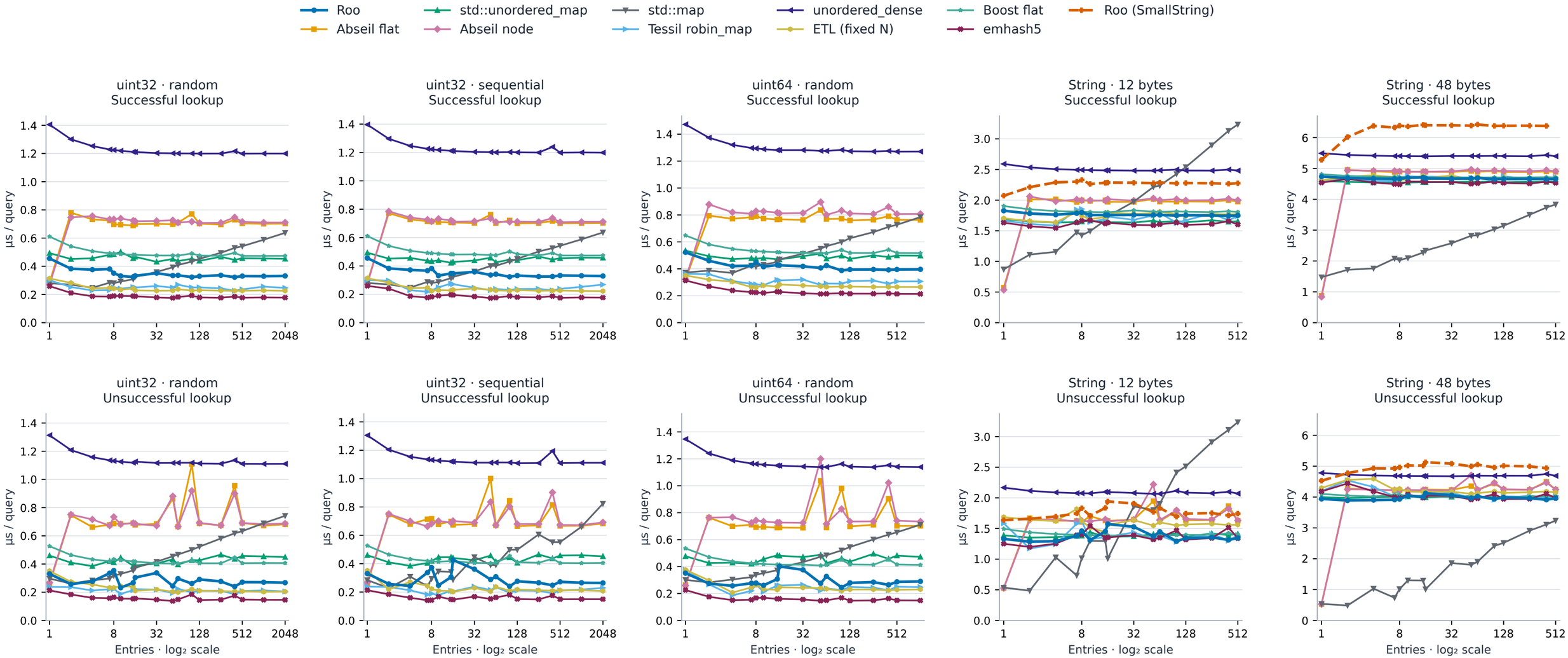
Medians shown; no confidence intervals. Default capacity policies; fill ratios are not normalized. Lower is better.

ETL: fixed capacity N, N buckets; embedded storage + adapter included; no dynamic-growth series.

Roo (SmallString): separate capture/board; other string series use std::string. 48-character SmallString ends at 400 entries.

Lookup performance across map sizes

Shared absl::Hash<Key> for every hash map; std::map uses ordering. Y axes are linear and start at zero.



Roo fill limit 75% · ESP32 · 240 MHz · -std=gnu++17 -O2 -DNDEBUG · warm access · heap checks: light · 1 suite/case, 7 samples/row

Medians shown; no confidence intervals. Default capacity policies; fill ratios are not normalized. Lower is better.

ETL: fixed capacity N, N buckets; embedded storage + adapter included; no dynamic-growth series.

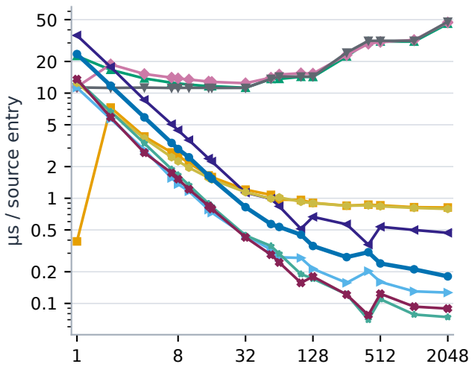
Roo (SmallString): separate capture/board; other string series use std::string. 48-character SmallString ends at 400 entries.

Copy and construction costs

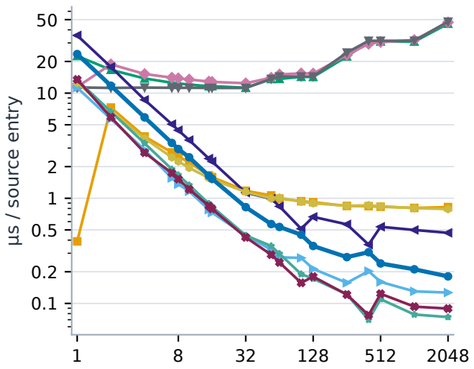
Shared absl::Hash<Key> for every hash map; std::map uses ordering. Y axes are logarithmic.

Roo
Abseil flat
std::unordered_map
Abseil node
std::map
Tessil robin_map
unordered_dense
ETL (fixed N)
Boost flat
emhash5
Roo (SmallString)

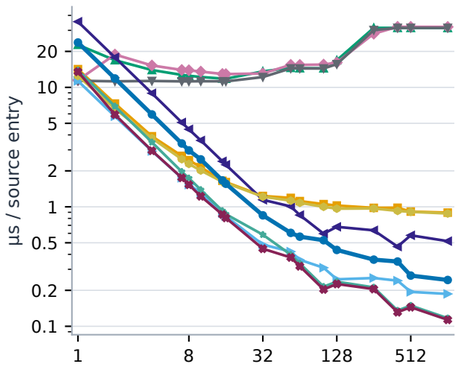
uint32 · random
Copy + destruction



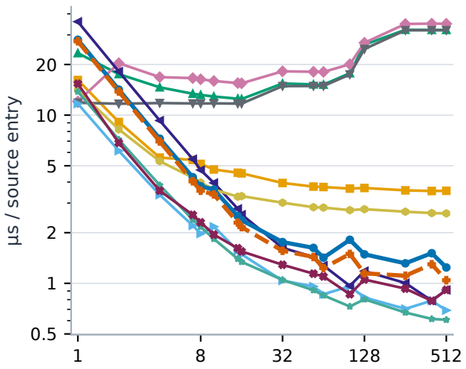
uint32 · sequential
Copy + destruction



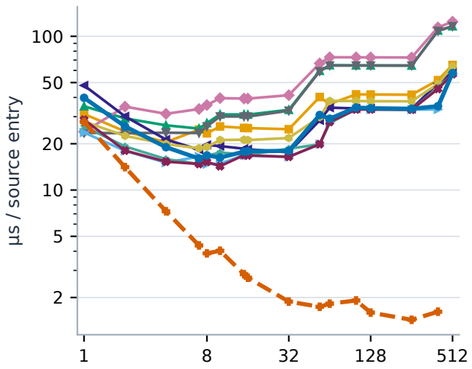
uint64 · random
Copy + destruction



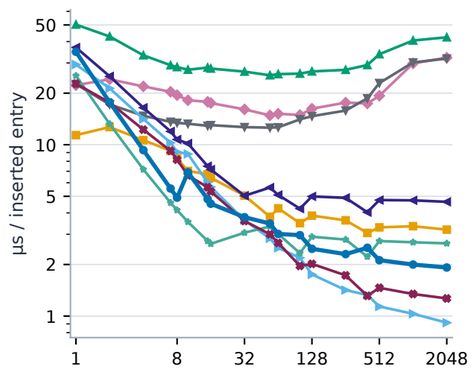
String · 12 bytes
Copy + destruction



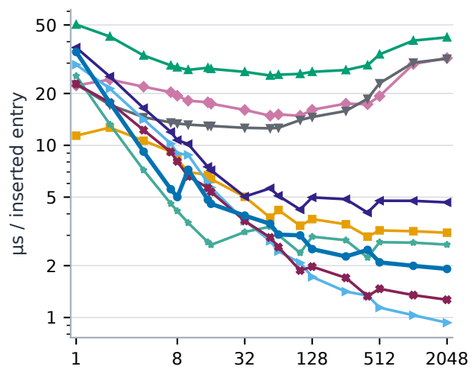
String · 48 bytes
Copy + destruction



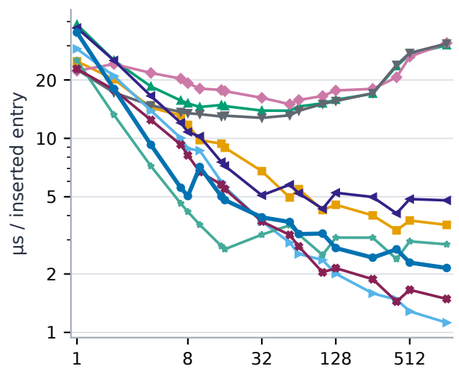
uint32 · random
Build with growth + destruction



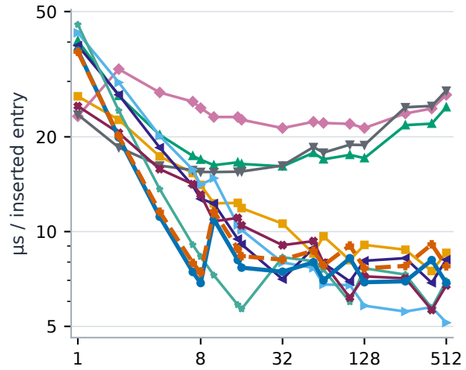
uint32 · sequential
Build with growth + destruction



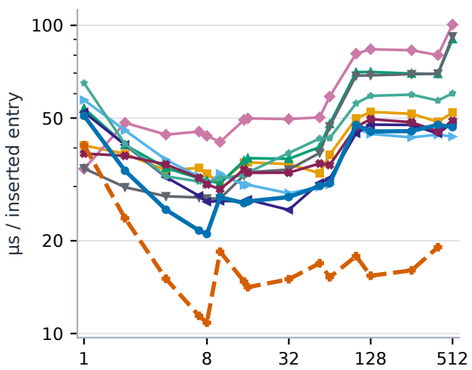
uint64 · random
Build with growth + destruction



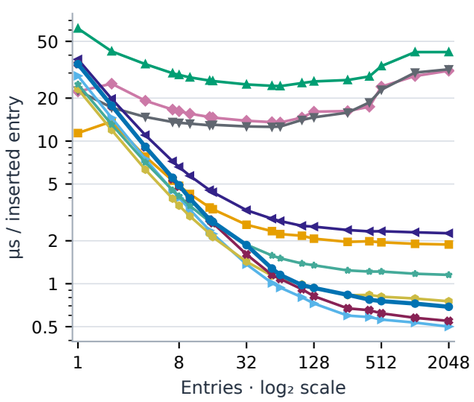
String · 12 bytes
Build with growth + destruction



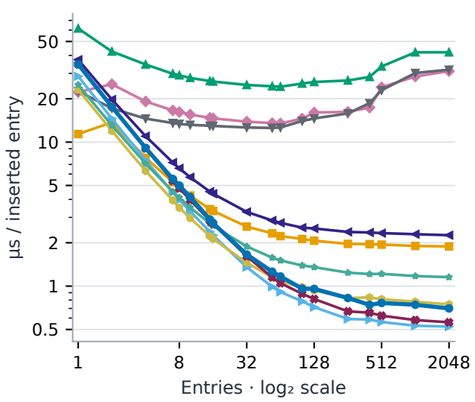
String · 48 bytes
Build with growth + destruction



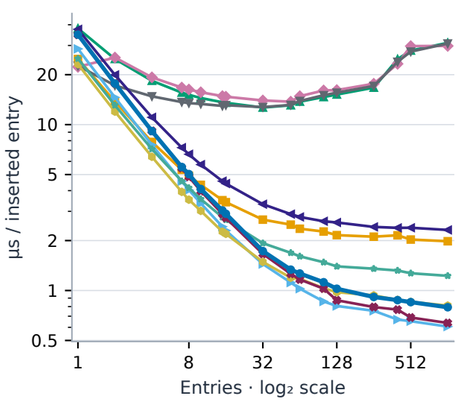
uint32 · random
Presized build + destruction



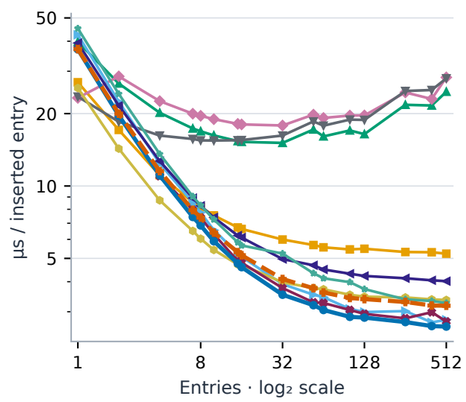
uint32 · sequential
Presized build + destruction



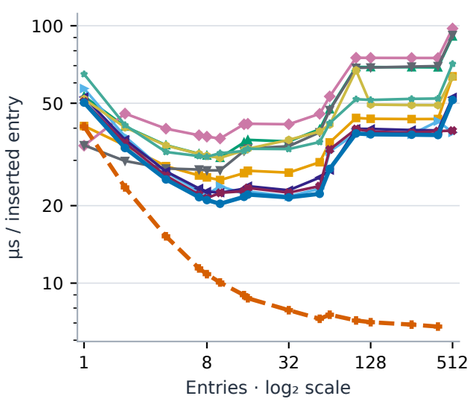
uint64 · random
Presized build + destruction



String · 12 bytes
Presized build + destruction



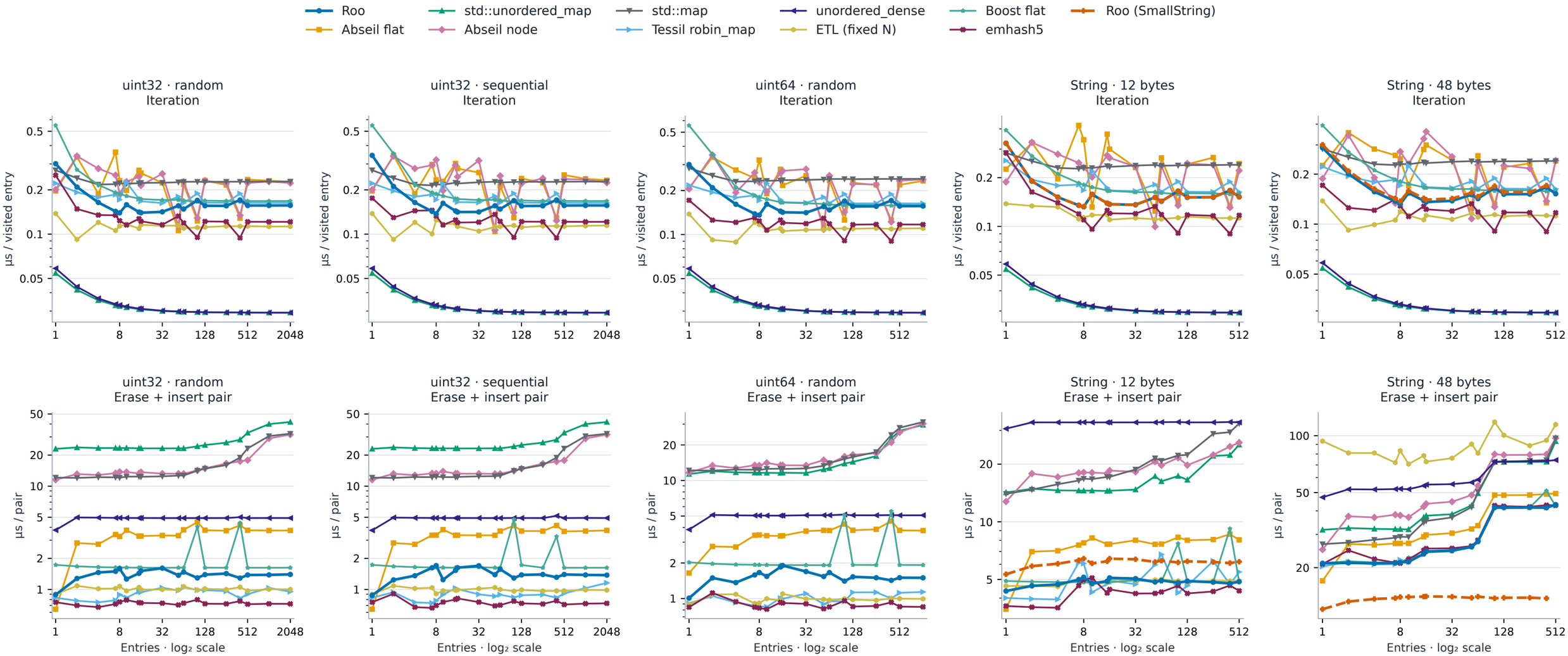
String · 48 bytes
Presized build + destruction



Roo fill limit 75% · ESP32 · 240 MHz · -std=gnu++17 -O2 -DNDEBUG · warm access · heap checks: light · 1 suite/case, 7 samples/row
Medians shown; no confidence intervals. Default capacity policies; fill ratios are not normalized. Lower is better.
ETL: fixed capacity N, N buckets; embedded storage + adapter included; no dynamic-growth series.
Roo (SmallString): separate capture/board; other string series use std::string. 48-character SmallString ends at 400 entries.

Iteration and erase/insert costs

Shared absl::Hash<Key> for every hash map; std::map uses ordering. Y axes are logarithmic.



Roo fill limit 75% · ESP32 · 240 MHz · -std=gnu++17 -O2 -DNDEBUG · warm access · heap checks: light · 1 suite/case, 7 samples/row

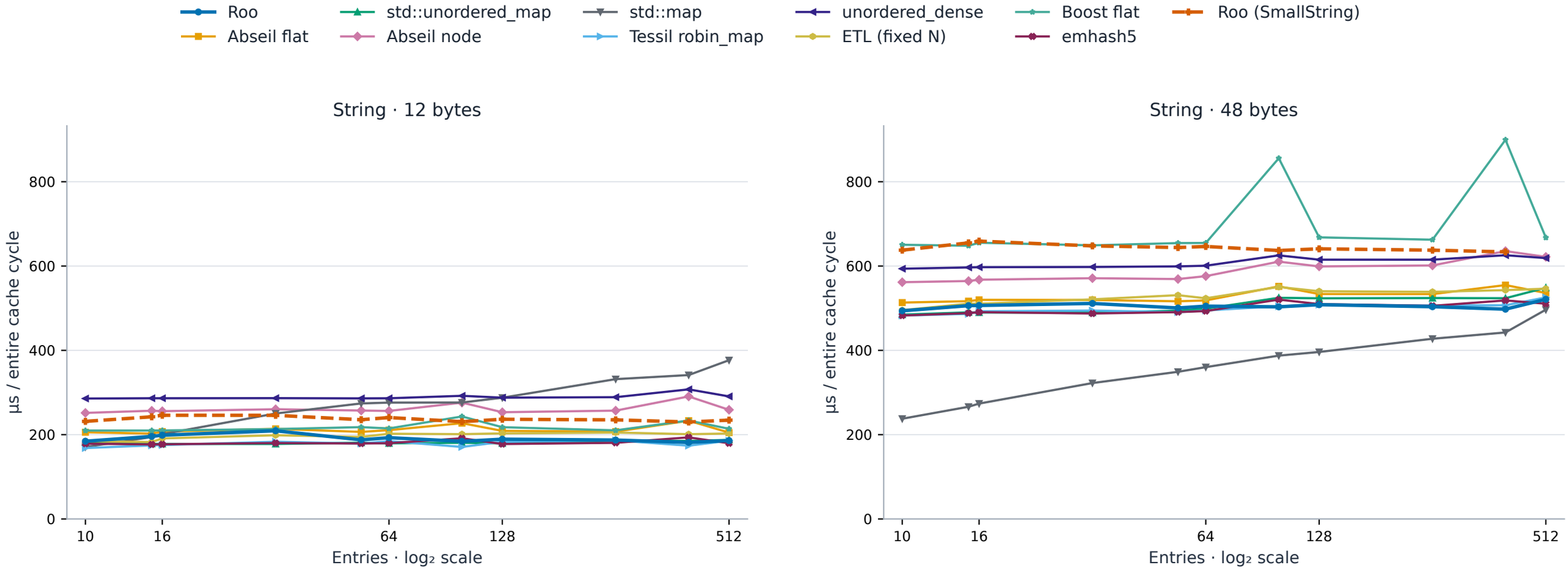
Medians shown; no confidence intervals. Default capacity policies; fill ratios are not normalized. Lower is better.

ETL: fixed capacity N, N buckets; embedded storage + adapter included; no dynamic-growth series.

Roo (SmallString): separate capture/board; other string series use std::string. 48-character SmallString ends at 400 entries.

Cache churn · string keys

Shared absl::Hash<Key>; std::map uses ordering. One cycle = 90 hits + 10 misses + one eviction + one insertion.



Roo fill limit 75% · ESP32 · 240 MHz · -std=gnu++17 -O2 -DNDEBUG · warm access · heap checks: light · 1 suite/case, 7 samples/row

Medians shown; no confidence intervals. Default capacity policies; fill ratios are not normalized. Lower is better.

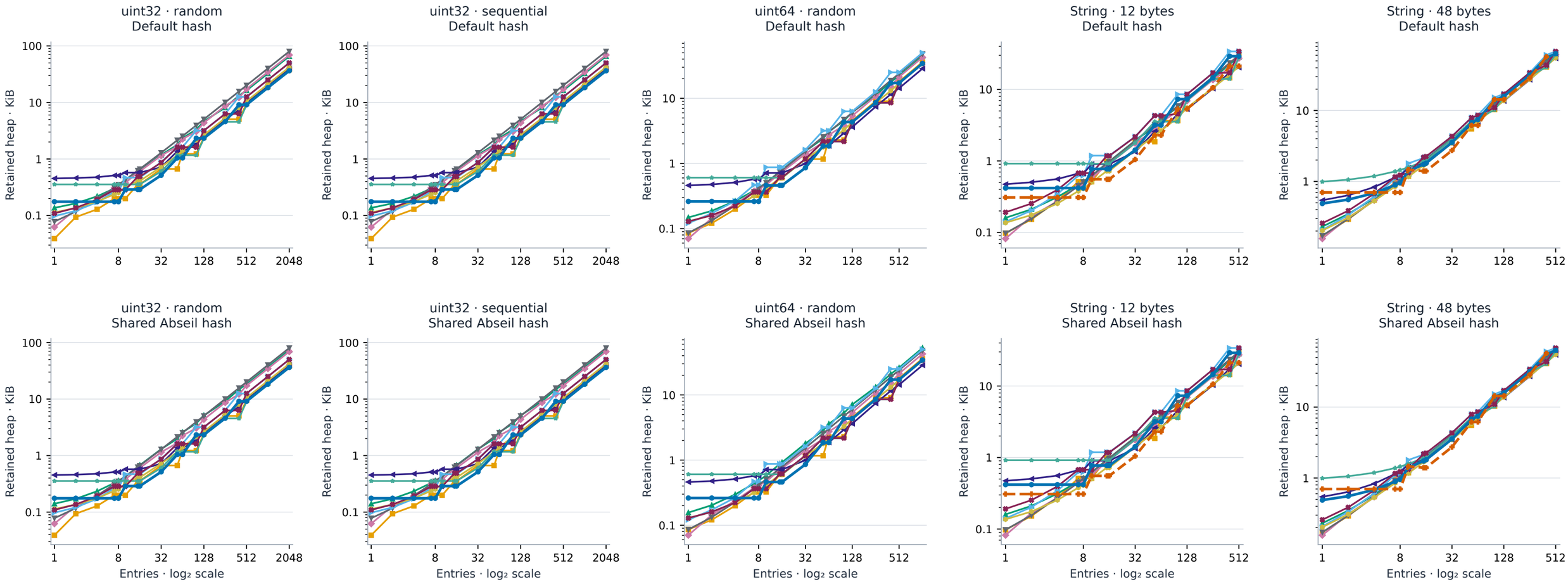
ETL: fixed capacity N, N buckets; embedded storage + adapter included; no dynamic-growth series.

Roo (SmallString): separate capture/board; other string series use std::string. 48-character SmallString ends at 400 entries.

Retained memory · default and shared hashing

Logarithmic axes. Grown-map footprint includes its object, owned strings and allocator overhead; not peak memory. std::map remains ordered.

Roo · Abseil flat · std::unordered_map · Abseil node · std::map · Tessil robin_map · unordered_dense · ETL (fixed N) · Boost flat · emhash5 · Roo (SmallString)



Roo fill limit 75% · ESP32 · 240 MHz · -std=gnu++17 -O2 -DNDEBUG · warm access · heap checks: light · 1 suite/case, 7 samples/row

Medians shown; no confidence intervals. Default capacity policies; fill ratios are not normalized. Lower is better.

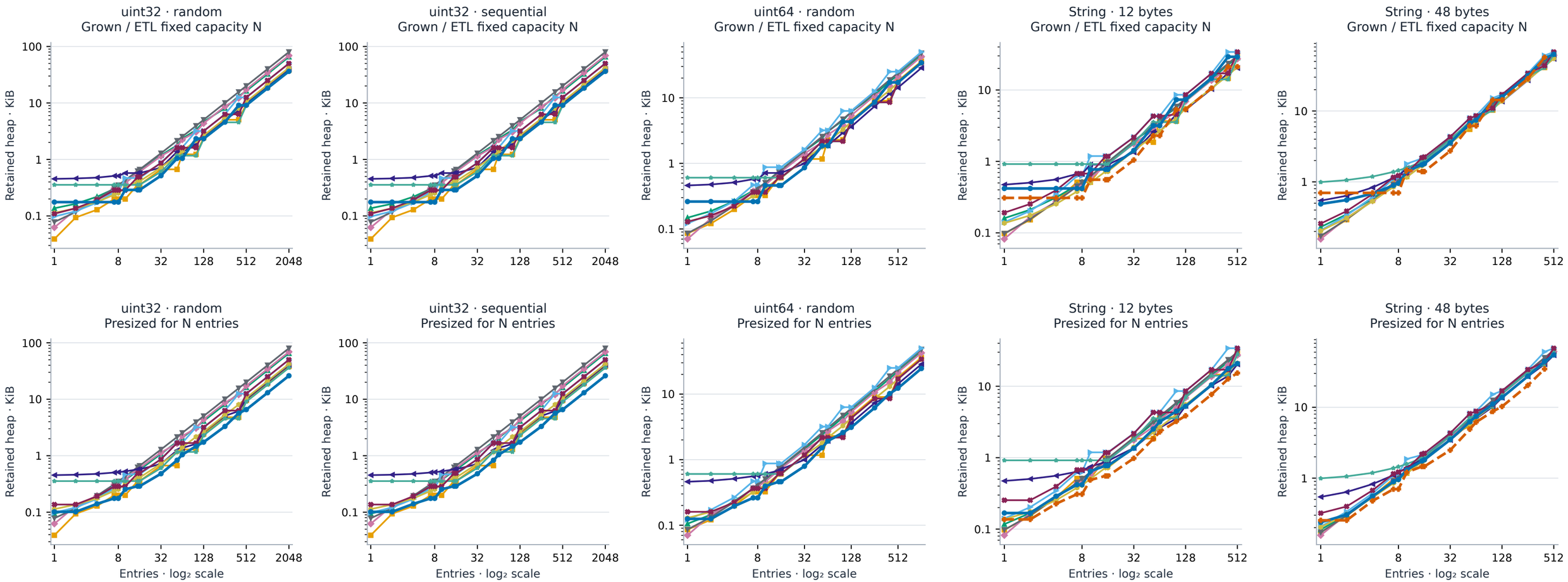
ETL: fixed capacity N, N buckets; embedded storage + adapter included; no dynamic-growth series.

Roo (SmallString): separate capture/board; other string series use std::string. 48-character SmallString ends at 400 entries.

Retained memory · growing versus presizing

Default hashers; logarithmic axes. Heap includes the map object, owned strings and allocator overhead; not peak memory. std::map stays ordered.

Roo · Abseil flat · std::unordered_map · Abseil node · std::map · Tessil robin_map · unordered_dense · ETL (fixed N) · Boost flat · emhash5 · Roo (SmallString)



Roo fill limit 75% · ESP32 · 240 MHz · -std=gnu++17 -O2 -DNDEBUG · warm access · heap checks: light · 1 suite/case, 7 samples/row

Medians shown; no confidence intervals. Default capacity policies; fill ratios are not normalized. Lower is better.

ETL: fixed capacity N, N buckets; embedded storage + adapter included; no dynamic-growth series.

Roo (SmallString): separate capture/board; other string series use std::string. 48-character SmallString ends at 400 entries.